Daniel Sainati

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Education

- **University of Pennsylvania** — *August 2024 - Ongoing*
  Doctor of Philosophy in Computer Science — Philadelphia, Pennsylvania
  Advisors: Benjamin C. Pierce and Stephanie Weirich
- **Cornell University** — *December 2017 - May 2018*
  Master of Engineering in Computer Science — Ithaca, New York
  Advisor: Adrian Sampson
- **Cornell University** — *August 2014 - December 2017*
  Bachelor of Science in Computer Science — Ithaca, New York

Employment

- **Senior Programming Language Engineer** — *October 2021 - July 2024*
  Flow Foundation — Vancouver, British Columbia
  - Worked on **Cadence**, a high level statically-typed smart contract language
- **Software Engineer** — *August 2018 - October 2021*
  Meta — Seattle, Washington
  - Worked on **Flow**, a static type system for Java Script, used at Meta on tens of millions of lines of code
- **Software Engineering Intern** — *Summer 2017*
  Meta — Menlo Park, California
  - Worked on **HHVM**, Meta's just-in-time Hack compiler
- **Software Engineering Intern** — *Summer 2016*
  Two Sigma Investments — New York, New York

Publications

Refereed Conference, Journal, and Workshop Publications

- **Modular GPU Programming with Typed Perspectives**. Manya Bansal, **Daniel Sainati**, Joseph W. Cutler, Saman Amarasinghe, and Jonathan Ragan-Kelley. In Programming Language Design and Implementation (PLDI) 2026 (**Distinguished Paper**). 25 pages.
- **The Search for Constrained Random Generators**. Harrison Goldstein, Hila Peleg, Cassia Torczon, **Daniel Sainati**, Leonidas Lampropoulos, and Benjamin C. Pierce. In Programming Language Design and Implementation (PLDI) 2026. 25 pages.
- **Typing Strictness**. **Daniel Sainati**, Joseph W. Cutler, Benjamin C. Pierce, and Stephanie Weirich. In Principles of Programming Languages (POPL) 2026. 31 pages.
- **LambdaLab: An Interactive λ-Calculus Reducer for Learning**. **Daniel Sainati** and Adrian Sampson. In SPLASH-E 2018. 10 pages.

Preprints

- Programming Languages and Formal Methods for GPUs. **Daniel Sainati**, Cynthia Richey, Paul Biberstein, Yuka Ikarashi, and Benjamin C. Pierce. Submitted for publication. 67 pages.

Books

- **Software Foundations in Lean**. Mike Hicks, Benjamin C. Pierce, and the **Software Foundations in Lean Team**.
  - Among the primary translators of the original Software Foundations into Lean across three volumes: *Logical Foundations in Lean*, *Hoare Logic in Lean*, and *Type Systems in Lean*

Honors, Funding, and Awards

- **NSF CSGrad4US Fellowship** *2023*
  National Science Foundation
- **eBay Undergraduate TA Award** *2016*
  Cornell University
- **Engineering Dean's List** *2014-2017*
  Cornell University

Talks

- **Modular GPU Programming with Typed Perspectives** *September 2026*
  Brown Cognitive Engineering Lab (invited) Providence, Rhode Island
- **Modular GPU Programming with Typed Perspectives** *September 2026*
  MathWorks Code Generation Knowledge Forum (invited) Natick, Massachusetts
- **Typing Strictness** *February 2026*
  New York University Formal Methods Seminar (invited) New York, New York
- **Typing Strictness** *February 2026*
  Portland State University PLV Seminar (invited) Portland, Oregon
- **Typing Strictness** *December 2025*
  Cornell Programming Languages Discussion Group (invited) Ithaca, New York
- **Typing Strictness and Laziness with Effects and Coeffects** *May 2025*
  New Jersey Programming Languages and Systems Seminar New York, New York

Teaching and Service

- **Co-organizer** *May 2026*
  New Jersey Programming Languages and Systems Seminar Philadelphia, Pennsylvania
- **Co-organizer** *2026*
  PLClub @ University of Pennsylvania Philadelphia, Pennsylvania
- **Student Volunteer**
  - January 2026 / POPL 2026 / Rennes, France
- **Teaching Assistant**
  University of Pennsylvania
  - Fall 2026 / CIS 5000 / Software Foundations
  - Fall 2025 / CIS 5000 / Software Foundations
- **Teaching Assistant**
  Cornell University
  - Spring 2018 / CS 6110 / Advanced Programming Languages
  - Fall 2017 / CS 4810 / Introduction to Theory of Computation
  - Spring 2017 / CS 3152 / Introduction to Computer Game Development
  - Fall 2015 - Fall 2016 / CS 3110 / Data Structures and Functional Programming (Fall 2016 Head TA)
  - Spring 2015 / CS 2110 / Object-Oriented Programming and Data Structures